

Introduction

Artificial intelligence continues to present unique opportunities and challenges across financial services, disrupting business models, investments and the workforce.

“ The rapid speed at which AI is maturing, combined with the explosion of potential use cases, presents a particular challenge for financial services executives and board members.

The digitalization of financial services – from the advent of ATMs (automated teller machines) and online banking to the widespread use of mobile apps – transformed the industry from all perspectives: providers, customers, employees, regulators and other stakeholders. Artificial intelligence (AI) is poised to have at least as profound an impact, albeit with important differences.

The digital era started in the mid-1980s and is broadly considered to have run for three to four decades. AI has, in fact, been around a lot longer; however, the introduction and widespread adoption of generative AI (genAI) and its increased accessibility two years ago have had a seismic effect on technology and business. Its potential to automate and augment tasks throughout the operating models of most organizations quickly established it as one of the most transformative technologies the world has ever seen. Banking, insurance, capital markets and payments are among the areas in which its impact is expected to be most pervasive (see Figure 1).

The rapid speed at which AI is maturing, however, combined with the explosion of potential use cases, presents a particular challenge for financial services executives and board members. Most have limited knowledge and experience of

technology, yet they are expected to develop a vision of how AI will change their industry, their business and operating models, their offerings and experiences, and their workforces. As some early adopters of genAI are already trumpeting their achievements, financial services leaders are starting to feel intense pressure to strategize their investments to seize the opportunities that are patently on offer. At the same time, executives will need to act immediately to defend their systems, data and customers against a potent new threat, while preparing to comply with a raft of new regulations.

Never before has the industry attempted to reinvent itself, its practices and its offerings so extensively at such a rapid pace. This paper aims to help industry leaders address this challenge by summarizing the areas in which AI is likely to have the greatest impact, the fundamental steps that are needed to achieve this impact, and the risks posed by the technology alongside how they might be mitigated. By examining trends and forecasts in the industry, the paper hopes to encourage engagement from all parties in the future role of AI in financial services. By promoting a better understanding of both the potential and the risks of the technology, it seeks to help build a foundation for the effective and responsible application of AI for the benefit of all.



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AI landscape in financial services

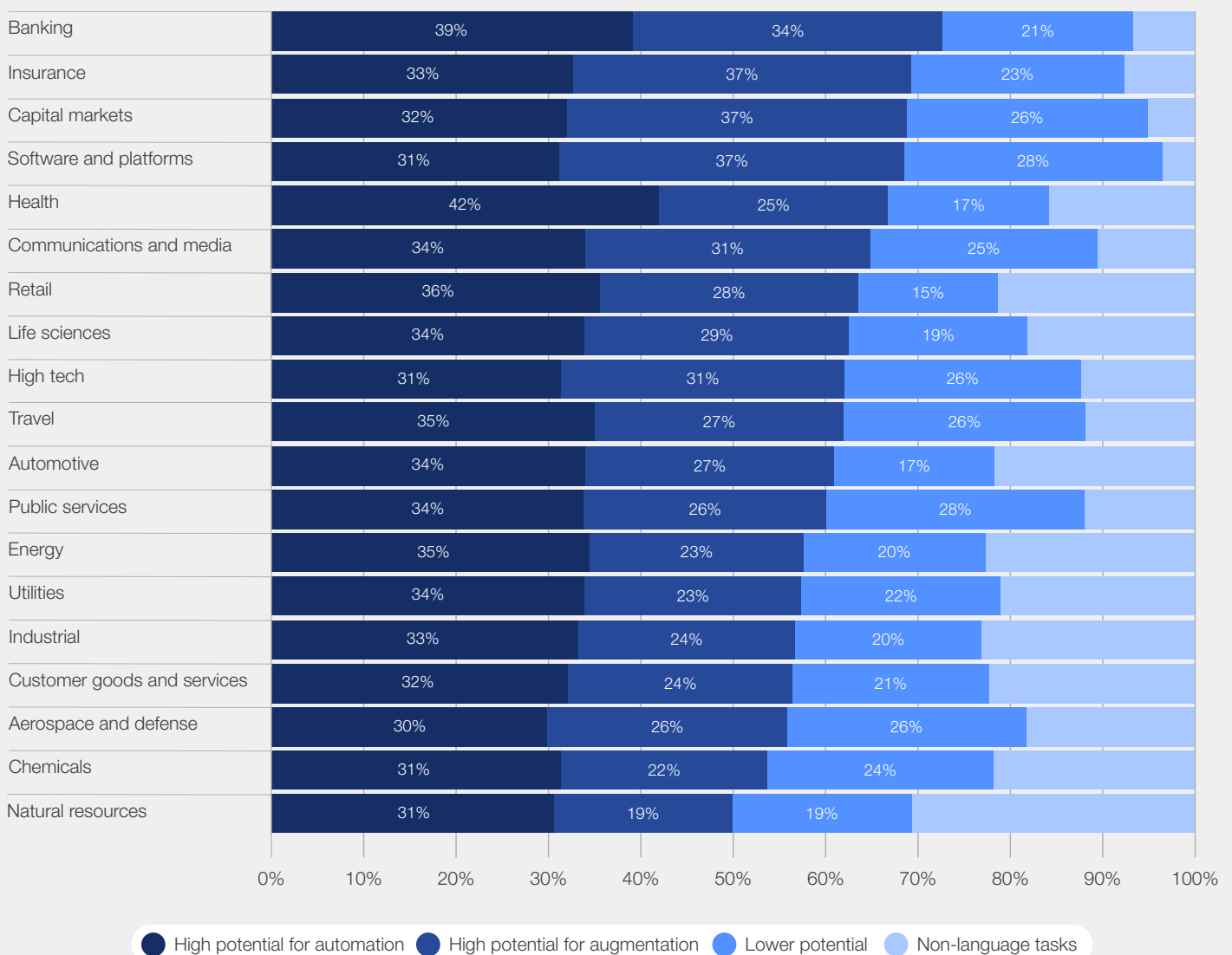
Financial services is leading in AI investment, harnessing the technology to enhance efficiency across functions, reduce risk and manage fraud.

Financial services firms have been using AI in their processes for decades. Shortly after the advent of genAI, it became evident that these firms – because so much of what they do is language-based – had a greater potential than most others

to capitalize on large language models (LLMs). In fact, the tasks that take up almost half of all the time spent by financial services employees are well suited to either automation or augmentation (see Figure 1).

FIGURE 1 The potential impact of LLMs across all major industries

Work time distribution across US industries



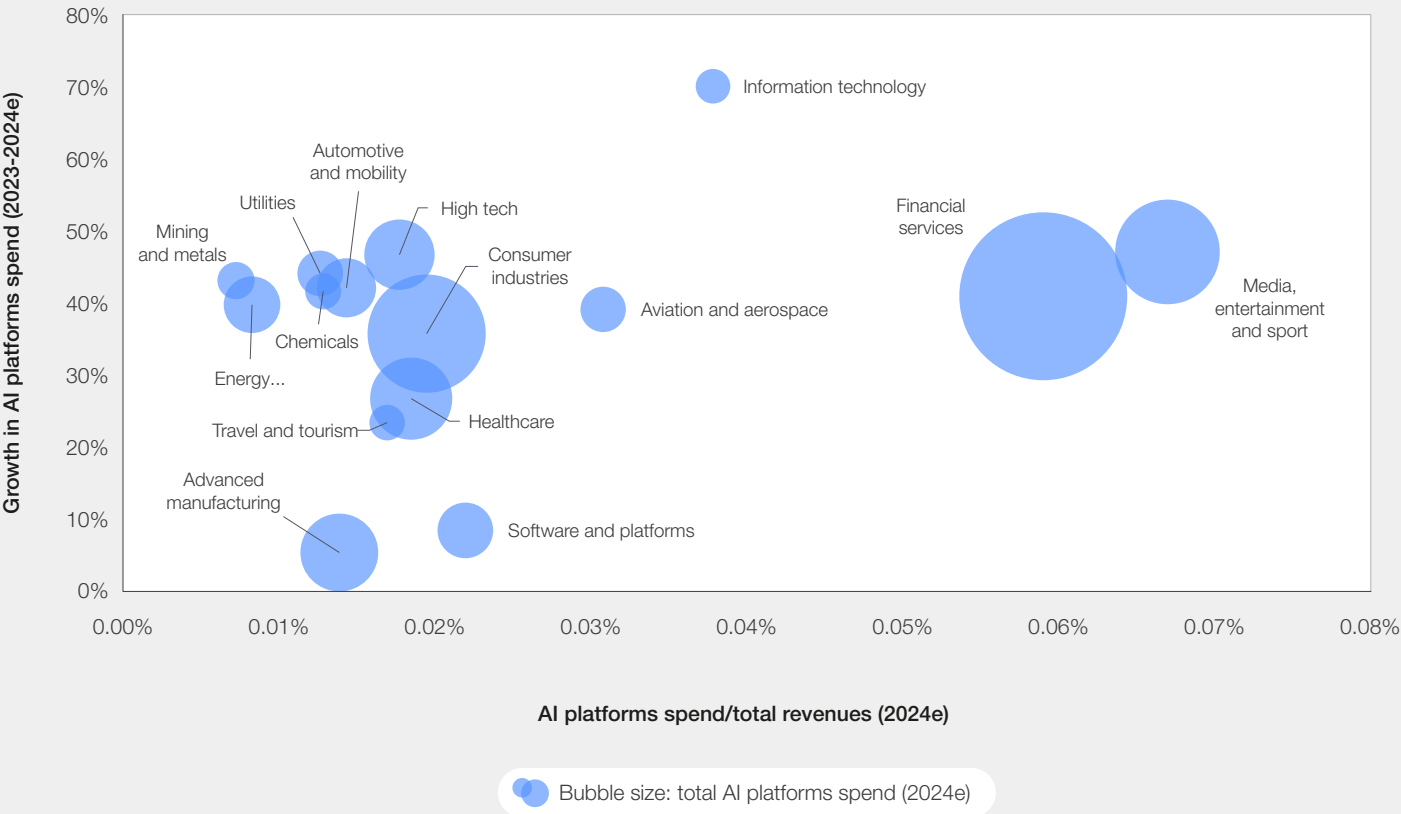
Source: Accenture. (2023.) A new era of generative AI for everyone. <https://www.accenture.com/content/dam/accenture/final/accenture-com/document/Accenture-A-New-Era-of-Generative-AI-for-Everyone.pdf#zoom=40>.

It's no surprise, therefore, that financial services businesses are exploring, investing in and implementing AI at a faster rate than other industries (see Figure 2). As its early pilots deliver promising outcomes and start to be scaled across the enterprise, the commitment of the industry is

only expected to deepen. Financial services firms spent a significant \$35 billion on AI in 2023, with projected investments across banking, insurance, capital markets and payments expected to reach \$97 billion by 2027.¹

FIGURE 2 **Estimated global AI spend, by industry (2024)**

AI platforms spend by industry, 2023-2024e
Growth in spend, spend/revenues and relative total spend



Source: Accenture Research. Data from Accenture's G2000 list. AI Platforms spend from IDC (2023-2024e). Revenues from Capital IQ S&P (2023-2024e)

As soon as financial services firms started to examine the potential of genAI, it became apparent that the technology could transform processes and practices all the way from back-office operations to front-end customer-facing experiences. Its ability to process and reference large amounts of data efficiently revealed numerous opportunities for automating routine manual tasks

in areas such as sales, customer and employee experiences, fraud and risk management, and technology development. In addition, companies are developing or acquiring a variety of AI-powered tools that help employees perform their current roles more efficiently and effectively. It remains to be seen whether there are any financial services tasks that will, in the long run, be unaffected by AI.





TABLE 1 | The most promising AI use case examples (contextualized) across financial services

Industry	Function	Description	Value delivered
Banking 	Sales and service	Customer service agents receive quick and comprehensive information on all aspects of products, policies and processes from a variety of sources	– Greater agent efficiency – Increased response accuracy – Quicker response time
Capital markets 	Client servicing/ investment management	Firms use AI models to create investment portfolios, offer financial assistance and provide clients with real-time insights and trading recommendations	– Enhanced client satisfaction and retention – Competitive advantage
Payments 	Fraud management and detection	Pre-emptive fraud detection includes technologies that can proactively seek and identify suspicious behaviour or anomalous events before fraudulent transactions ²	– Improved fraud protection for customers – Enhanced customer experience by minimizing false positives
Insurance 	Claims	The automation of claims and customer document processing ³	– Improved workflows – Greater agent efficiency – Streamlined document collection and validation
Across financial services 	Risk management and underwriting	Prediction of fraudulent transactions, more effective underwriting processing and risk scoring	– Reduced internal and external risk – Better protection of data – Improved underwriting processing times – Greater accessibility to established credit scoring and evaluation
	Technology development	Streamlining the software development life cycle, from writing code to automation testing ⁴ as well as understanding and decommissioning of legacy code environments	– Improved workflow and accuracy – Increased efficiency – Shorter development cycles – Reduction in technology debt

Seeing early value from AI implementation

After achieving success from efficiency opportunities, businesses are developing future strategies backed by lessons learned and recurring themes.

70%

of financial services executives believe that AI will directly tie to revenue growth in upcoming years.

It is clear from the above that there are countless potential applications of AI within the operating models of financial services firms; however, most surveys indicate that the primary objective today is efficiency. This is understandable, as cost-reduction programmes tend to deliver quicker and more directly measurable results than growth or risk management programmes. Dramatic outcomes go a long way towards building support for continued investment. Nonetheless, 70% of financial services executives believe that AI will directly tie to revenue growth in upcoming years⁵ by enhancing customer experiences, making products and services more personalized and relevant, enabling innovative new offerings, empowering cross- and up-selling, and improving security against potential threats.

There are, of course, many ways in which AI can enable these improvements. As financial services leaders continue to identify and prioritize new use cases and develop their investment strategies, however, there are common themes that frequently recur:

- **Personalized customer experiences:** AI-powered virtual assistants can provide thorough, tailored, round-the-clock support for customers' more basic issues (e.g. product recommendations or applications); for complex enquiries, they can augment the capabilities of human agents, allowing for quicker, more relevant responses (e.g. questions around investment decisions and portfolio allocation adjustments).
- **Product innovation and revenue growth:** genAI enables financial services firms to address market niches, such as mass-affluent advisory services, more effectively than was previously feasible. It is also enabling innovation and the creation of new revenue streams (e.g. by combining synthetic customer data with more efficient A/B testing for new deposit and lending products). Through its ability to process and synthesize data, it can also identify and help realize opportunities for up- and cross-selling.
- **Data-driven decisions:** The power of AI to convert large amounts of data into useful, tailored insights allows the technology to enhance investment decisions, improve risk assessments and generally help business leaders make better day-to-day decisions – allowing leaders to spot key trends (e.g. in transaction volumes, customer feedback or market trends).
- **Improved risk management, compliance and security:** AI can scrutinize data, transactions and other events more thoroughly, quickly and tirelessly than humans can. It is being used to constantly monitor for cybersecurity threats and to identify suspicious activity in real time. It can also be hugely helpful in relieving data collection and reporting burdens – especially during the mandatory identification process of customers, known as KYC (know your customer).



Moving towards an AI-powered future

Over the next decade AI will revolutionize the industry through enhanced customer experiences, support and decision-making capabilities.

Thanks largely to the growing use of AI and data, and the value companies are starting to generate from their AI strategies, financial services businesses will look very different a decade from now. With further platform digitalization, smarter automation and more interactive decision-based workflows, we can expect to see significant changes in how customers bank, invest, borrow and insure assets. The services they use will become more accessible, efficient and personalized. The benefits for customers fall mainly into two categories:

- **Enhanced customer experiences:** Firms will use AI to synthesize massive amounts of data. This will include financial data (e.g. bank

accounts, investment accounts and insurance policies) and other customer information (e.g. key moments such as the first purchase of a home). This will significantly improve the efficiency of most customers' interactions with their banks. It will also enable more relevant advice that helps customers plan effectively for retirement, major purchases or even healthcare-related expenses.

- **Improved customer support:** By using smart AI agent chatbots, customers could receive audio and video responses to their questions. AI agents can action custom requests without human intervention, providing quicker and more accurate solutions.



“ Executives must continually evaluate technology ecosystems to capture emerging opportunities, ensuring that AI investments are thoughtfully integrated into broader initiatives.

To achieve these improvements and keep up with the rapid pace of innovation, business leaders will require a flexible AI strategy. They will also need to continuously monitor the technology landscape, especially over the next two to three years, as developments advance in key areas:

- **Small language models (SLMs):** An SLM is a machine-learning algorithm that has been trained on a dataset that is smaller and more specific than those of large language models (LLMs). SLMs are typically deployed for a single specific task (e.g. answering customers’ questions about a particular product); they can be more efficient and faster than LLMs due to their size and their use of a more targeted dataset.
- **Retrieval-augmented generation (RAG):** RAG is a process that enhances the capability of genAI models by optimizing LLMs – increasing both accuracy and reliability. This process uses in-house data repositories (e.g. internal organizational data such as policy or procedural documents) to support the validation of responses to ensure it is combining the latest and most accurate responses.⁶ Companies can incorporate RAG into their AI-based assistants, making them more effective and reliable at answering customer questions, by validating responses against a master repository (such as knowledge articles in a repository).
- **AI agents:** These are designed and built to understand inputs, make decisions and act on

them without any human intervention.⁷ When AI acts as an agent instead of a service, it is no longer locked into assisting with a single task or function, but can be applied to a broader range of problems and decision-making opportunities. An example of this is the use of AI agents that can process enquiries, action customer requests and make product recommendations.⁸

- **Quantum computing:** By integrating quantum computers with conventional supercomputers, parts of traditional problem-solving workflows can be accelerated. Leveraging quantum computing in AI models would allow faster processing of larger datasets and accelerated decision-making, especially in areas requiring complex pattern recognition. An example of this would be quicker identification of fraudulent transactions for banks and payment processors.⁹

As AI becomes central to technology strategies, executives must continually evaluate technology ecosystems to capture emerging opportunities, ensuring that AI investments are thoughtfully integrated into broader initiatives. Decisions by traditional financial institutions on partnerships or investments in financial technology companies – which are re-emerging as leaders in niche AI-driven solutions – will grow in importance.¹⁰ Striking the right balance between managing investment costs and risks, accelerating time to market and selectively building in-house solutions will be key to progressing towards scaled adoption of AI in financial services.

AI in the workforce

For AI strategies to succeed, companies must prioritize talent development and reskilling, while cultivating a culture that enables seamless collaboration between humans and technology.

90%

of leaders believe their organization needs to consider significant adjustments or total transformation of their reskilling strategy to support the future.

It is broadly acknowledged that for AI to achieve its full potential, especially in the future, more is required than just the technology and the data that fuels it. The role of people is also crucial. People are needed to develop, implement, collaborate with and in some cases operate the systems. Also, as genAI will soon be as ubiquitous as Microsoft's operating systems and Google's browser are today, it is vital that all employees embrace the technology and learn how to use it optimally. Further research will be needed to better understand employee attitudes towards AI and related workplace dynamics, to allow executives to plan the best path forward. An important step towards a future where humans and AI collaborate at scale is acknowledging workers' concerns about redundancy and changes to their roles, and communicating the firm's AI strategy openly and credibly.

As technology redefines existing roles, creates new ones and transforms how work is performed, it will fundamentally change the culture of organizations and mindset of its people. Leaders are already

starting to see the impacts of AI in their workforce – 90% believe their organization needs to consider significant adjustments or total transformation of their reskilling strategy to support the future.¹¹

AI cannot fulfil expectations if the implementation strategy does not adequately include talent and culture.

There is much to be learned. Financial services firms will be competing with all others for the limited number of experienced AI technicians and will need to grow much of their own talent. They have also realized that entirely new skills are emerging – such as prompt engineer – and they will need to train for those too. All employees, at all levels, will require ongoing training as new work tools are introduced and upgraded. With technology playing an ever-greater role in the organization, senior executives will also need to acquire a thorough understanding of its current and potential contributions if they are to develop and execute successful business strategies.^{12,13}



AI risks and challenges

AI still poses great risks to financial services with, increasing concerns of misinformation, deepfakes and data privacy.

“ One of the most significant risks of using AI in financial services is misinformation, leading to potential market manipulation and fraudulent transactions.

AI is eliciting much enthusiasm among industry leaders; yet this is not without reservations. In addition to workforce challenges, there are potential negative impacts emerging from the deployment of the technology. These may be deliberate, where AI is employed as a weapon by cyber assailants or other malefactors, or inadvertent, caused by well-intentioned users (e.g. by introducing unintended biases into underwriting decisions). Given the tremendous capability of most AI tools, the potential for harm is real, especially in industries like financial services or healthcare, where customer privacy and trust are critical to success.

This accounts for the slower deployment of AI in areas such as the customer experience, where cyber threats, market manipulation, and data bias

and privacy concerns pose significant challenges. As companies become more familiar with customer-facing AI use cases, however, they are also becoming more comfortable with managing these risks.¹⁴

There are various risks associated with using AI in financial services; one of the most significant is the spread of misinformation, leading to potential market manipulation and fraudulent transactions, which is an increasing concern across all industries. Figure 4 highlights that the risk of misinformation and disinformation is ranked first in the short term (up 15 positions from the previous year) and fifth in the longer term, based on a cross-industry perception survey of more than 1,500 experts who were asked about major global risk concerns two and 10 years from now.¹⁵

FIGURE 4 Global risks ranked by severity over the short and long terms (industry agnostic)

2 years

1		Misinformation
2		Extreme weather events
3		Societal polarization
4		Cyber insecurity
5		Interstate armed conflict
6		Lack of economic opportunity
7		Inflation
8		Involuntary migration
9		Economic downturn
10		Pollution

10 years

1		Extreme weather events
2		Critical change to Earth systems
3		Biodiversity loss and ecosystem collapse
4		Natural resource shortages
5		Misinformation and disinformation
6		Adverse outcomes of AI technologies
7		Involuntary migration
8		Cyber insecurity
9		Societal polarization
10		Pollution

● Economic
 ● Environmental
 ● Geopolitical
 ● Societal
 ● Technological

Source: World Economic Forum. (2024). These are the 3 biggest emerging risks the world is facing. <https://www.weforum.org/agenda/2024/01/ai-disinformation-global-risks/>.

“ Another serious, and related, threat is the ability of genAI to create and disseminate synthetic (fabricated or deepfake) content.

Another serious, and related, threat is the ability of genAI to create and disseminate synthetic (fabricated or deepfake) content. This risk has grown as both the ability and accessibility of genAI have increased, as evidenced by a 223% spike in deepfake-related tool trading on dark web forums in the first quarter of 2024, compared to the same period in 2023.¹⁶ Plausible images, video and voices can be created with remarkable ease to impersonate familiar figures to induce fraudulent transactions or other behaviour. For example, a finance worker at a multinational firm was summoned on to a video call, ostensibly by his chief financial officer and other members of staff, and was instructed to transfer \$25 million to an account controlled by fraudsters.¹⁷ As is often the case with consumers who are duped by deepfakes of family members, the finance worker did not suspect that the chief financial officer and other colleagues on the call were computer generated and part of an elaborate scam. It doesn't take much imagination to appreciate how incidents such as this could negatively impact financial markets and the global economy.

Although the above risks, and others, have been exacerbated by AI, the technology is also being used to detect and/or prevent threats. To an increasing degree, AI is providing:

- **Authentication technology:** Uses digital watermarks or metadata that are embedded in new content at the time of creation to help prove its authenticity.¹⁸
- **Detection technology:** Scours a vast landscape of activity to identify potential threats. It is also able to act, autonomously in some cases, to nullify detected threats. For example, it can recognize fake content without needing to compare it to the original,¹⁹ and it can swiftly examine the code within an application to determine whether its purpose is benign or malicious.²⁰



Prioritizing responsible AI

Executives must address self-governance, cybersecurity, privacy and ethical concerns as they build out AI initiatives.

84%

of financial organizations are implementing or planning a framework to govern how AI will be built, trained, used and audited.

Any organization seeking to benefit from AI will need to address a host of critical issues up-front; self-governance, fraud and cybersecurity, data privacy, workforce management and the spread of disinformation are only the most immediate. This is especially relevant in a highly regulated area such as financial services, where firms serve collectively as a cornerstone of the global economy and a bulwark that protects individual customers and businesses against those who threaten their financial well-being.

Responsible AI is a broad term that covers these and other issues. It refers to the practice of designing, building and deploying AI in a manner that empowers employees and businesses and impacts customers and society fairly. There are four main pillars:

- **Organizational:** Democratizing the new way of working and facilitating human-machine collaboration.
- **Operational:** Setting up governance and systems that will enable AI to flourish.
- **Technical:** Helping ensure systems and platforms are trustworthy and explainable by design.
- **Reputational:** Articulating the firm's AI mission and ensuring this is anchored to its values, ethical guardrails and accountability structure.

As companies implement AI more extensively, with ever-increasing sophistication and decision-making responsibility, it will be easy to lose sight of this imperative. Imperfections such as a minor demographic bias can quickly have an outsized effect when the technology is scaled across the firm's operating model. To achieve responsible AI, leaders should leave no doubt that this is a corporate priority, and should disseminate a set of practical guidelines that help to embed it in people's daily actions and the corporate mindset. As they begin to define their AI ecosystems, they will also need to evaluate their vendors, and the grouping of multiple AI systems based on their vision of the role of AI in their future.

Businesses across financial services continue to prioritize responsible AI, especially in areas like self-governance, with 84% of financial organizations implementing or planning a framework to govern how AI will be built, trained, used and audited, to adhere to business principles and relevant regulations.²¹ Alongside this, there is growing support for external groups and coalitions, like the Fintech Open-Source Foundation (FINOS) and the Taiwan Financial Supervisory Commission. These bodies work together to support AI policy development, collaboration and innovation while also promoting the safe and effective use of AI across financial services.



AI regulation challenges

As AI regulation continues to evolve, there's a need to balance the pace of technology innovation against many other factors.

“ The regulation of AI will evolve as business leaders, AI researchers, policy-makers and others continue to grapple with these issues and work towards resolving gaps in current policy.

Progress has been made in AI policy and regulation; however, the topic remains a priority for business leaders. Countries and regions are working to establish a standard set of policies, with various groups beginning to define and publish frameworks. Examples include the Monetary Authority of Singapore (MAS) publishing a set of AI methodologies,²² the European Union's AI Act,²³ and the US President Biden's AI Executive Order.²⁴ Concerns persist in financial services, however, about the inconsistent approaches of regulators around the world, particularly the EU's more comprehensive AI Act compared to President Biden's AI Executive Order in the US.

Despite ongoing efforts there are three issues that have yet to be resolved regarding current AI policy. These continue to cause uncertainty across financial services.

1. **The pace of innovation:** The speed at which AI is evolving and being implemented presents a challenge for businesses and regulatory bodies that lack the resources to keep abreast of the implications and emerging threats.
2. **What to regulate:** A related challenge is caused by the proliferation of AI models and capabilities; identifying which should be scrutinized and

regulated is a mammoth task. As one size seldom fits all models, policy-makers will need to focus on the relevant use cases (e.g. objectivities, data models, responses) in order to implement successful regulatory measures.²⁵

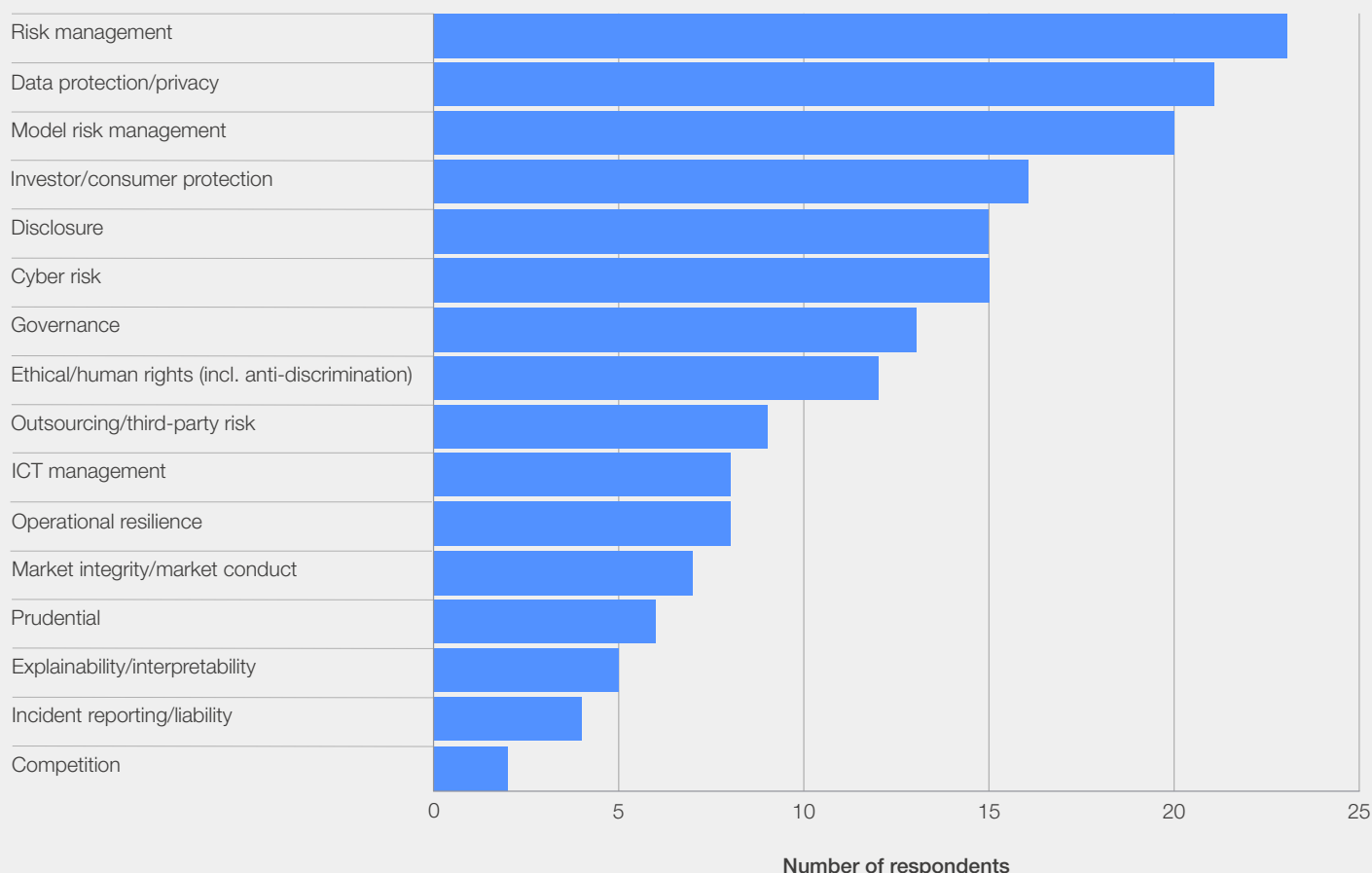
3. **Who should regulate:** The various parties have yet to agree on the appropriate level of oversight for AI policy – industry, state, federal or regional – as well as how variations of policy should be managed, especially globally.²⁶

Though these questions remain and are presently impacting the pace of new AI regulation, some industry leaders and policy-makers have made the argument that financial services, due to its existing regulations, is better positioned than others when it comes to AI technology. A survey of 49 constituents from the Organisation for Economic Co-operation and Development (OECD), made up of various global members with experience in policy, economics and industry, highlights to some extent that current regulation is appropriate for governing the majority of AI uses. Examples of this can be found in areas such as consumer and investor protection, model risk management, third-party risk management, disclosure requirements, IT governance, cybersecurity, and fairness (see Figure 5 for additional examples).²⁷



FIGURE 5 | Examples of areas covered by existing financial services rules

Share of 49 responding OECD and non-OECD members provided non-exhaustive examples of rules and regulations that may apply to the use of AI financial services.



Note: The OECD brings together member countries and a range of partners that collaborate on key global issues at national, regional and local levels ensuring governments can work together to share experiences and seeks solutions to common problems – using OECD’s data and analysis to inform policy decisions.

Source: Organisation for Economic Co-operation and Development (OECD). (2024). *Regulatory approaches to Artificial Intelligence in finance*. <https://doi.org/10.1787/f1498c02-en>.

Although these existing regulations do not specifically mention AI, respondents agree that the underlying principles apply regardless of the technology. There is a level of agreement that current policies cover many AI uses; however, there is a call for more standardized guidance to reassure business leaders and technology advisors that new regulations will not undermine their substantial AI investments.

Some industry commentators hold the conflicting view that comprehensive regulation will impede the adoption of AI by banks and insurers, leaving them at a competitive disadvantage to somewhat less regulated players in the industry – such as fintechs

and non-bank lenders – resulting in greater risk to customers and the global economy.

The regulation of AI will evolve as business leaders, AI researchers, policy-makers and others continue to grapple with these issues and work towards resolving gaps in current policy. They will need to be clear and consistent about the purpose of new regulations, and to keep a wary eye open for unintended consequences. They will also need to maintain a careful balance as they seek to protect private and public interests while simultaneously promoting innovation and investment across all industries.^{28,29}